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Intelligent optimal design for near-zero energy buildings performance: A case study of five climate zones in China

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ABSTRACT

To enrich the methods used to analyze the sustainable performance of near zero energy consumption buildings (NZEBs), this paper introduces a hybrid intelligent algorithm combining the peak hedgehog optimizer (CPO), natural gradient boost (NGBoost) algorithm, and multi-objective gray wolf optimizer (MOGWO). Meanwhile, a knowledge and data-driven analysis framework is constructed. By using building information modeling-design builder (BIM-DB) simulation software and Monte Carlo simulation (MCS), a building data sample set of five climate regions in China is obtained. The input and output indices are determined according to a dual knowledge–data–driven framework, and a performance prediction model is constructed on the basis of the CPO–NGBoost machine learning algorithm. The objective function is determined by the relationship between the input and output indices evaluated by the prediction model, and the Pareto frontier is derived using the MOGWO. Finally, the optimal scheme of the multiobjective optimization problem of building performance is obtained by using the ideal point method. Using a case study, we found that (1) the parameters and thresholds for the building are defined using knowledge-driven approaches. Through BIM–DB simulation and Monte Carlo simulation, a dataset of architectural samples can be effectively generated. (2) The CPO–NGBoost machine learning algorithm has good predictive performance. The R^2 of all the prediction targets exceeds 0.9. (3) The CPO–NGBoost–MOGWO hybrid algorithm optimizes the building energy consumption to 17.59 %–34.25 % in five climate zones in China and optimizes the other performance metrics of NZEBs. Our approach can inform the sustainable performance of NZEBs in different climatic regions worldwide.

Nomenclature

x_1	Building orientation, °	BOR
x_2	External wall heat transfer coefficient, W/(m^2·K)	EWHTC
x_3	Exterior wall solar radiation absorption coefficient	EWSAC
x_4	Roof heat transfer coefficient, W/(m^2·K)	RHTC
x_5	Roof solar radiation absorption coefficient	RSAC
x_6	Air tightness, ac/h	AT
x_7	Window-wall ratio	WWR
x_8	Outer window heat transfer coefficient, W/(m^2·K)	OWHTC
x_9	Outer window solar radiation absorption coefficient	OWSAC
x_{10}	Outer window transmittance	OWT
x_{11}	Personnel density, person/m^2	PD
x_{12}	Air conditioning cooling setting temperature, °C	ACCT
x_{13}	Air conditioning heating setting temperature, °C	ACHT
f_1	Building energy consumption, MWh	BEC
f_2	Carbon emission, ton	CE
f_3	Thermal comfort	PMV
f_4	Glare index	GI

1. Introduction

1.1. Current trends and process

The swift expansion of the construction sector has led to a surge in greenhouse gas emissions, intensifying global warming and highlighting the urgency of addressing sustainability [1]. Due to the swift progression

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of urban development and increasing social requirements, energy consumption in the construction sector has significantly risen over the past few decades [2]. At present, the overall energy consumption in the building sector in China primarily comes from public buildings, residential structures, and industrial facilities [3]. Public buildings represent the largest portion of energy usage, which has grown five times over the last ten years [4]. Therefore, the energy efficiency of public buildings must be improved to conserve energy and reduce emissions. Near-zero-energy-consumption buildings (NZEBs) are buildings that consume almost no nonrenewable energy, instead consuming mostly renewable energy. In this context, NZEBs are effective for promoting environmental protection and achieving global carbon emission targets. In the process of realizing NZEBs, reasonable building envelope parameters can greatly decrease energy usage in buildings and enhance the comfort of residents [5].

Methods for obtaining accurate raw data are important for NZEBs design research [6]. The traditional method of acquiring effective data is to detect buildings in real time. Building data were obtained from a 23–story tower through monitoring, and simulations were carried out using EnergyPlus to suggest viable strategies for low-carbon adaptation [7]. Zhou investigated the building data of an NZEB to investigate its operational performance [8]. Traditional methods have long data acquisition cycles and require data of a certain scale. Through the application of intelligent concepts in the field of building construction, building data can be quickly obtained [9]. To date, building information modeling (BIM) technology combined with simulation software can quickly obtain effective datasets [10]. Uddin used BIM modeling to investigate the effects of the interior layout on the energy-saving behaviors of occupants [11]. Sha developed an integrated building energy system automation and optimization framework based on BIM [12]. The DesignBuilder (DB) relationship is characterized by a comprehensive software platform that relies on visual building modules and authoritative databases. DB simulates the overall energy usage of buildings through numerical simulation, providing support for subsequent architectural design optimization [13]. In addition, on the basis of the original dataset obtained by BIM–DB, an effective approach is required to derive the fitness function for multi-objective optimization challenges.

In recent years, machine learning (ML), which can handle large amounts of data and is widely used in various fields, has provided the possibility for building performance prediction models and obtaining fitness functions in multiobjective optimization problems [14]. Among these technologies, artificial neural networks (ANNs) [15] and support vector machines (SVMs) [16] are relatively common ML algorithms. Artificial intelligence techniques enable the analysis of larger datasets through the integration of intricate linear combinations. Nonetheless, the structural designs of ANNs are complex, and their application is limited [17]. Due to the scarcity of available data and the kernel function of SVMs, selecting appropriate hyperparameters is quite challenging; thus, the prediction model cannot achieve the best prediction effect [18]. Natural gradient boosting (NGBoost) is a new boosting algorithm that solves the problem of probability density distribution prediction by introducing a natural gradient [19]. The natural gradient has the beneficial property of keeping the parameters unchanged under different distribution changes so that the optimization problem is not affected by parameterization. The NGBoost algorithm has been applied to models in many fields. Ran analyzed the relationship between SHARP parameters and continuous solar flares using NGBoost [20]. Mitrentsis used NGBoost to construct a highly accurate, reliable, and clarified photovoltaic power generation prediction model [21]. Consequently, utilizing the NGBoost method to develop a performance prediction model is particularly appropriate. The high-precision building performance forecasting model developed using the NGBoost algorithm can accurately predict building performance in a short time, thereby reducing both the duration and cost of building simulations. In addition, compared with traditional empirical equations, the NGBoost proxy model can accurately and quickly obtain complex relationships between parameters and building performance [22]. By employing the NGBoost proxy model as a fitness function, the multi-objective optimization challenge can be addressed more rapidly and efficiently, thus decreasing both optimization duration and costs [23].

In NGBoost, hyperparameters are important factors affecting the performance characteristics of machine learning algorithms. Choosing suitable hyperparameters can significantly enhance the capabilities of algorithms, thereby increasing the model's accuracy and robustness [24]. The selection of inappropriate hyperparameters may lead to overfitting or underfitting problems and increase the resources and cost of model training [25]. Therefore, before the NGBoost prediction model is established, it is necessary to select its hyperparameters effectively to improve the accuracy, robustness, and generalizability of the model prediction. The integration of the crested porcupine optimizer (CPO) and the NGBoost model enables rapid identification of the model's hyperparameters while ensuring accurate predictions.

CPO can adjust the search area adaptively in the search process, improve the search efficiency, reduce unnecessary calculations, and reduce the weight of the prediction model [26]. Compared with traditional grid search (GS) [27] and random search (RS) [28], CPO can search the hyperparameter space more comprehensively and accurately and obtain a better combination of hyperparameters. The CPO algorithm is extensively applied in the hyperparameter optimization of machine learning (ML) algorithms, as it not only effectively addresses high-dimensional optimization challenges but also achieves high accuracy [26]. Fan demonstrated that CPO has significantly improved search efficiency and accuracy with an increasing number of hyperparameters [29]. Gao used the CPO–LSTM model for the prediction of bridge temperature [30]. Li predicted the carbon dioxide concentration with CPO as a hyperparameter optimization algorithm and BiLSTM as a prediction model [31]. To achieve multiobjective prediction and optimization, the CPO algorithm is applied to the hyperparameter optimization of the NGBoost model. Additionally, CPO–NGBoost is utilized to develop the building performance prediction model and to create the fitness function for multiobjective optimization of building performance.

Building energy consumption (BEC) [32], thermal comfort (PMV) [33], and carbon emission (CE) [34] are the key output indicators in the research field of NZEBs. On the basis of existing studies, we consider the glare index (GI) as an output index to comprehensively consider the sustainable performance characteristics of NZEBs [35,36]. Data analysis provides scientific support for NZEB research. However, the information extracted from the data is not interpretable. Therefore, we establish a dual-driven analysis model of knowledge data, integrate the lack of information and domain knowledge extracted from the data as a single method, and improve the accuracy and interpretability of the model. The knowledge and data-driven approach is a comprehensive research methodology aimed at integrating domain expertise and extensive data analysis to enhance research quality and scientific rigor. In this approach, the knowledge-driven aspect relies primarily on expert experience, theoretical knowledge, and industry standards, whereas the data-driven aspect is focused on extracting feature patterns from data results and conducting data analyses, such as simulation analysis. By combining knowledge-driven approaches, prediction processes and results become increasingly interpretable, especially when multiple target complex systems are involved, and the involvement of knowledge increases the transparency and reliability of the model. For instance, during the examination of building energy usage statistics, established patterns of energy consumption (such as higher usage in winter compared to summer) are employed to detect anomalous data, thereby ensuring the integrity of the input data for the model.

Numerous optimization techniques have been utilized within the architecture domain. Common algorithms include the second-generation nondominated sorting genetic algorithm (NSGA-II) [37] and particle swarm optimization (PSO) [38]. However, the existing methods have difficulty satisfying three optimization objectives, mainly because of the performance limitations of the optimization algorithms.

The multiobjective gray wolf optimization algorithm (MOGWO) is an advanced algorithm that combines the gray wolf optimization algorithm (GWO) and multiobjective optimization theory. MOGWO integrates exploration and exploitation strategies while leveraging the principles of collective intelligence. This approach helps prevent premature convergence to local optima and enhances the model's overall search capability and solution efficiency, making it well-suited for large-scale optimization challenges. Dilip used MOGWO to remodel complex electric systems [39]. Makhadmeh, S. N. reviewed the successful application of MOGWO in multiple fields, such as scheduling systems, network systems, and the cloud computing environment [40]. Zhang adopted MOGWO to solve the problem of multiobjective energy savings and low carbon, with coupling between wind energy and power [41]. In conclusion, the utilization of the MOGWO algorithm for multiobjective optimization in building performance can enhance various performance indicators, achieve an optimal equilibrium between BEC, CE, PMV, and GI, and ultimately improve the overall benefits of buildings.

Prior to conducting simulation analysis, the simulation parameters must be determined. As passive optimization techniques, the building envelope and building orientation play important roles in reducing the thermal load of NZEBs [36,42]. As a form of active optimization, an air conditioning system has a great influence on the BEC [43,44]. In addition, the BEC is affected by personnel-related parameters [45]. In this paper, the air conditioning system, building orientation, enclosure structure, personnel density, and other parameters are selected as parameters that affect building performance.

According to the distribution characteristics of climate factors, China can be divided into five climatic regions: warm-winter and hot-summer regions, mild regions, cold-winter and hot-summer regions, cold regions, and extremely cold regions. Owing to geographical differences, the measures to achieve NZEBs change according to the climate [46]. Drawing from extensive simulation data, Xu employed an artificial neural network (ANN) to forecast the yearly energy consumption of NZEBs in Shenyang, a city located in a cold region of China. [47]. Wu used the MOGWO algorithm for the five-objective optimization of NZEBs in China [48]. Yu analyzed and evaluated the influences of energy conservation strategies on the power consumption of NZEBs [49]. Various research directions for NZEBs in specific climate regions exist, including energy consumption prediction, effective measure exploration [50], energy consumption target optimization, and policy evaluation [51]. However, few researchers have analyzed the influences of different climate conditions on energy conservation by public buildings in a country. Owing to different climatic conditions, parameters such as building envelopes and air conditioning operations in one climatic zone may not be applicable in other climatic zones. Thus, the influences of different climate conditions on building energy consumption must be explored to save energy and reduce carbon emissions [52]. In this paper, a breakthrough is made regarding these research objects. All five climate regions in China are selected as test points for simulation and data analysis, demonstrating the method's thoroughness and applicability in this work.

1.2. Existing questions and solutions

The main questions addressed in this paper are as follows.

(1) China encompasses five climatic zones, each characterized by distinct building parameters. How can we utilize building simulation to obtain valid data for these diverse climatic zones?
(2) There exists a complex interdependence between building performance and the parameters that influence it. When constructing building performance prediction models, how can we overcome the limitations of traditional machine learning algorithms, such as their inability to effectively handle high-dimensional data and their limited model generalization capabilities, when dealing with such intricate relationships?
(3) BEC, CE, PMV, and GI are the four most critical indicators in the realm of building sustainable performance. How can we further employ optimization algorithms to achieve performance enhancements across multiple dimensions and obtain optimal building design solutions?

The innovation points and solutions proposed in this paper are as follows.

(1) Under the framework of dual-knowledge data-driven approach, key parameters and appropriate thresholds are established. Building models are created using BIM software and imported into DB software in the gbXML format for simulation analysis. The integration of simulation results from BIM-DB and the Monte Carlo Simulation (MCS) method can generate accurate building performance indicators. This combination facilitates the acquisition of building simulation data, making it more convenient and efficient, and reduces the complexity of data collection and processing. This method effectively addresses the high data volume demand in building performance analysis and provides a theoretical and data foundation for subsequent research.
(2) A CPO-NGBoost performance prediction model has been developed to address the issue of insufficient prediction accuracy in traditional models due to the low accuracy of hyperparameter selection. By employing CPO for multi-objective optimization of NGBoost's hyperparameters, the adaptability and generalization of the prediction model in handling high-dimensional data are enhanced. Meanwhile, this approach improves the lightweight computation of the prediction model, thereby reducing the time cost of model analysis.
(3) The MOGWO algorithm is selected, and the analysis results of the CPO-NGBoost prediction model are used to determine the objective function. The chosen multi-objective optimization algorithm, MOGWO, simulates the hunting behavior of gray wolves. In intricate multi-objective optimization scenarios, the group search mechanism can effectively avoid local optima and quickly find the global Pareto front solutions. When dealing with the four-objective optimization of building sustainable performance, including BEC, CE, PMV, and GI, this method has the advantages of high efficiency, stability, and diversity. It provides personalized optimization solutions for building performance improvement under different climatic conditions, thus supporting sustainability within the construction sector and fostering innovative research methodologies.

The structure of the remainder of this paper is outlined as follows. In Section 2, the theoretical foundations of the CPO-NGBoost-MOGWO algorithm are presented, establishing a theoretical framework for the subsequent research. Section 3 details the process of constructing and processing datasets within a knowledge-data dual-driven framework using BIM-DB, as well as the augmentation of datasets using Latin Hypercube Sampling (LHS). It also explains the development of building performance prediction models for five regions in China using CPO-NGBoost and the establishment of a multi-objective optimization model using the MOGWO algorithm to investigate sustainable building performance across different regions. Section 4, building on the framework and procedures outlined in Section 3, provides a detailed modeling and analysis of specific cases, discussing the application and computational results of the proposed CPO-NGBoost-MOGWO algorithm. To verify the advancement and superiority of the proposed research method, Section 5 compares it with existing methods and discusses the impact of building parameters on sustainable development in different regions. Finally, Section 6 summarizes all the conclusions drawn in this paper and outlines directions for future work.

Table 1
CPO-NGBoost-MOGWO algorithm pseudocode.

#Algorithm: CPO-NGBoost performs multi-objective prediction

Import: X: predicted target matrix (BEC, CE, PMV, and GI); t: variable iteration times to determine the number of cycles; T_{max}: maximum number of function evaluation times;
Output: $f\left(\overrightarrow{x_i^t}\right) : f_{BCE}\left(\overrightarrow{x_i^t}\right)/f_{CE}\left(\overrightarrow{x_i^t}\right)/f_{PMV}\left(\overrightarrow{x_i^t}\right)/f_{GI}\left(\overrightarrow{x_i^t}\right)$
1: Initialize the input parameters and evaluate the fitness value of $\overrightarrow{x_{CP}^t}$
2: **While** $t < T_{max}$
3: Update $\overrightarrow{y_i^t}$ based on formula (6), and update population size N based on formula (4);
4: **For** $i = 1{:}N$ do
Building and training NGBoost models
5: Hyperparameter selection (*colsample_bytree*, *learning_rate*, *n_estimators* and *minibatch_frac*)
6: Update the defense mechanism
7: **If** $f\left(\overrightarrow{x_i^t}\right) > f\left(\overrightarrow{x_i^{t+1}}\right)$ $\overrightarrow{x_i^t} = \overrightarrow{x_i^{t+1}}$
End If
8: **End For**
9: **End While**
#Algorithm: MOGWO performs multi-objective prediction
11: Initialize the gray wolf population $X_i(i = 1, 2, \ldots, n)$, determine the objective function $f_i(j)$.
12: Create an archived population and calculate the non-dominant solution.
13: $t = 1$, update the population position to avoid selecting the same leader.
14: **While** ($t <$ Max number of iterations)
For update population location
End For If the archive is full. Run the network mechanism to update the solution set
End If
$t = t{+}1$
End While
15: Get multi-objective optimization solution.

2. Preliminaries

2.1. Natural gradient boosting (NGBoost)

NGBoost comprises three modular components that must be pre-configured: the base learner (f), the parameter probability distribution (P_θ), and the relevant scoring rule (S). For a new sample input x, the predicted result $y|x$ is provided in the form P_θ. Parameter θ is obtained via the superposition of M base learner inputs and an initial distribution $\theta^{(0)}$. To obtain the predicted result parameter θ, each base learner f takes x as input, and the predicted output is scaled with a scaling factor and learning rate; the updated formula is shown in Formula (1) [21]:

$$y\Big|x \sim P_\theta(x), \theta = \theta^{(0)} - \eta\sum_{m=1}^{M}\rho^{(m)} \cdot f^{(m)}(x) \tag{1}$$

where ρ is the scaling factor; the scoring rule S is the gradient of parameter θ on the probability distribution P_θ, denoted by $\nabla S(\theta, y)$. This direction rises the fastest. Shifting a very small number of parameters in this gradient direction can increase most scoring rules, as shown in Formula (2) [19]:

$$\widetilde{\nabla} S(\theta, y) \propto \lim_{\varepsilon \to 0} \underset{d \,:\, D_S(P_\theta \| P_{\theta+d}) = \varepsilon}{\arg\ max} S(\theta + d, y) \tag{2}$$

where d is the change in the gradient direction and D is the divergence.

By addressing the aforementioned optimization problem, the natural gradient can be derived as follows:

$$\widetilde{\nabla} S(\theta, y) \propto I_S(\theta)^{-1} \nabla S(\theta, y) \tag{3}$$

where $I_S(\theta)$ is the Riemannian measure of a statistical manifold at θ derived from the scoring rule S.

2.2. Crested porcupine optimizer (CPO)

CPO is an innovative metaheuristic algorithm introduced in 2023, drawing inspiration from the defensive behaviors exhibited by crested porcupines. In response to threats, porcupines employ four defensive strategies: visual, auditory, olfactory, and physical aggression. The CPO uses these four mechanisms for exploration and development. The newly introduced cyclic population reduction technique enhances algorithm performance, accelerates convergence, and increases diversity, thereby preventing the algorithm from becoming stuck in local minima. The parameters N′, T_{max}, α, T_f, T, and N_{min} are established, and the population is initialized randomly. When $t < T_{max}$, the fitness scores of the candidate solutions are assessed to identify the optimal solution.

$$N = N_{min} + (N' - N_{min}) \times \left(1 - \left(\frac{t\%\frac{T_{max}}{T}}{\frac{T_{max}}{T}}\right)\right) \tag{4}$$

$$\overrightarrow{x_i^{t+1}} = \overrightarrow{x_i^t} + y \times \left|2 \times \tau_2 \times \overrightarrow{x_{CP}^t} \times \overrightarrow{y_i^t}\right| \tag{5}$$

$$\overrightarrow{y_i^t} = \frac{\overrightarrow{x_i^t} + \overrightarrow{x_r^t}}{2} \tag{6}$$

$$\overrightarrow{x_i^{t+1}} = \left(1 - \vec{U}_1\right)\overrightarrow{x_i^t} + \vec{U}_1 \times \left(\vec{y} + \tau_3 \times \left(\overrightarrow{x_{r1}^t} \times \overrightarrow{x_{r2}^t}\right)\right) \tag{7}$$

$$\overrightarrow{x_i^{t+1}} = \left(1 - \vec{U}_1\right)\overrightarrow{x_i^t} + \vec{U}_1 \times \left(\overrightarrow{x_{r1}^t} + S_t^i \times \left(\overrightarrow{x_{r2}^t} - \overrightarrow{x_{r3}^t}\right) - \tau_3 \times \vec{\delta} \times \gamma_t \times S_i^t\right) \tag{8}$$

$$\overrightarrow{x_i^{t+1}} = \overrightarrow{x_{CP}^t} + (\alpha(1 - \tau_4) + \tau_4) \times \left(\delta \times \overrightarrow{x_{CP}^t} - \overrightarrow{x_i^t}\right) - \tau_5 \times \vec{\delta} \times \gamma_t \times \overrightarrow{F_i^t} \tag{7a}$$

$$S_t^i = \exp\left(\frac{f(x_t^i)}{\sum_{k=1}^{N} f(x_t^k) + \psi}\right) \tag{9}$$

$$\overrightarrow{F_i^t} = \overrightarrow{\tau_i} \times \frac{m_i \times \left(\overrightarrow{x_i^{t+1}} - \overrightarrow{x_i^t}\right)}{\Delta t} \tag{10}$$

$$m_i = e\frac{f\left(\overrightarrow{x_i^t}\right)}{\sum_{k=1}^{N} f\left(\overrightarrow{x_k^t}\right) + \psi} \tag{11}$$

where $\overrightarrow{x_{CP}^t}$ denotes the best solution obtained and represents CP. $\overrightarrow{y_i^t}$ in-

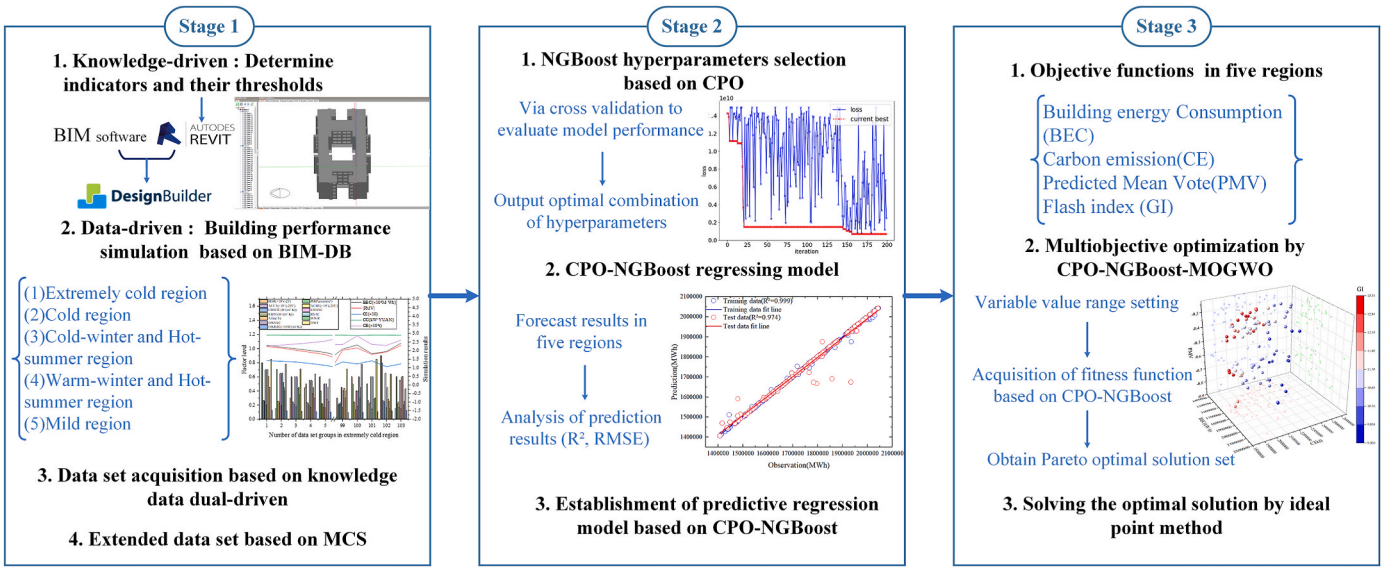


Fig. 1. Flow chart of multi-objective optimization based on CPO-NGBoost-MOGWO hybrid algorithm.

dicates the vector formed between the current CP and a randomly selected CP from the population, reflecting the position of the predator at iteration *t*. $\vec{U}_1$ represents a binary vector within the range. S_t^i denotes the scent diffusion factor; the values of this factor range from 0.3 to 2.6, and its size determines the location of the randomly selected solution being developed. $\vec{\delta}$ indicates a parameter used to control the search direction. τ_i is a random number in the interval [0, 1]. $\overrightarrow{x_i^t}$ refers to the position of the i-th individual at iteration t and the predator at that position. $\overrightarrow{F_i^t}$ signifies the average force affecting the *i*-th predator's CP, and γ_i is a random number within [1, N]. $\overrightarrow{x_i^t}$ is the initial speed of the *i*-th individual at the *t*-th iteration, and $\overrightarrow{x_i^{t+1}}$ is the final speed of the *i*-th individual at the next iteration, that is, the (*t*+1)-th iteration.

2.3. Multi-objective gray wolf optimizer (MOGWO)

MOGWO is a multi-objective optimization algorithm based on the gray wolf optimizer (GWO), which simulates the social behaviors and hunting mechanisms of gray wolves. This algorithm addresses conflicting optimization objectives by incorporating two main components: the archive and the leadership selection mechanism. Moreover, this algorithm combines multi-objective optimization with swarm intelligence, and its adaptability and flexibility in complex environments allow it to effectively address specific problems and identify the Pareto-optimal solution set.

The population parameters of the wolf pack are initialized, fast nondominated sorting is performed, and the archive population is established to obtain the most suitable nondominated solutions through position updates. The method for calculating the crowding distance of each individual is presented as follows:

$$D(i) = \sum_{i=1}^{n} \frac{f_i(j+1) - f_i(j-1)}{f_{i\max} - f_{i\min}} \tag{12}$$

where $f_i(j+1) - f_i(j-1)$ represents the difference between the (*j*-1)th function and the (*j*+1)th function within the objective function, and $f_{i\max}$ and $f_{i\min}$ denote the maximum and minimum values of the objective function, respectively.

The second crucial element entails utilizing a roulette wheel method to choose nondominated solutions from the sparsest regions of the archive to serve as leaders. The probability of selection is inversely related to population density, facilitating the avoidance of local optima and improving optimization performance.

$$p_i = \frac{c}{M_i} \tag{13}$$

where *c* is a constant (c > 1) and M_i is the number of *i*-th Pareto-optimal solutions.

Table 1 shows the CPO-NGBoost-MOGWO algorithm pseudocode.

2.4. Shapley Additive Explanations (SHAP)

The SHAP method has consistently interprets of the influences of characteristic variables on the prediction results well. SHAP can be employed to examine the impacts of characteristic variables on the prediction outcomes, as well as to assess both the positive and negative influences these variables have on the results. The SHAP technique calculates Shapley values for different feature variables to evaluate their effects on the predicted outcomes. The interpretation function is represented by Formula (14) [53]:

$$g(z') = \phi_0 + \sum_{i=1}^{N} \phi_i z_i', \phi_i \in R \tag{14}$$

where $g(\cdot)$ denotes the interpretation function, ϕ_0 represents the Shapley value of the featureless virtual model, *N* signifies the number of input features in the sample, ϕ_i denotes the Shapley value of the i-th feature, and z' ($z' \subset \{0,1\}^N$) indicates the presence or absence of each characteristic. The impact of each feature is determined using Formula (15) [54]:

$$\phi_i = \sum_{S \subseteq F \setminus \{x_i\}} \frac{|S|!(N - |S| - 1)!}{|F|!} [f(S \cup \{x_i\}) - f(S)] \tag{15}$$

where x_i represents the eigenvector, *S* denotes all subsets of features that exclude feature x_i, $|S|$ indicates the count of nonzero elements in *S*, and *F* encompasses the entire set of all features in the model.

3. Methodology

Fig. 1 illustrates the multi-objective optimization flow chart that utilizes the CPO–NGBoost–MOGWO hybrid algorithm. The input and output indicators and their thresholds are determined through a dual knowledge–data-driven framework. On the basis of the parameter design of the orthogonal experiment, BIM–DB is used to obtain simulation data of the building performance, and the Monte Carlo simulation

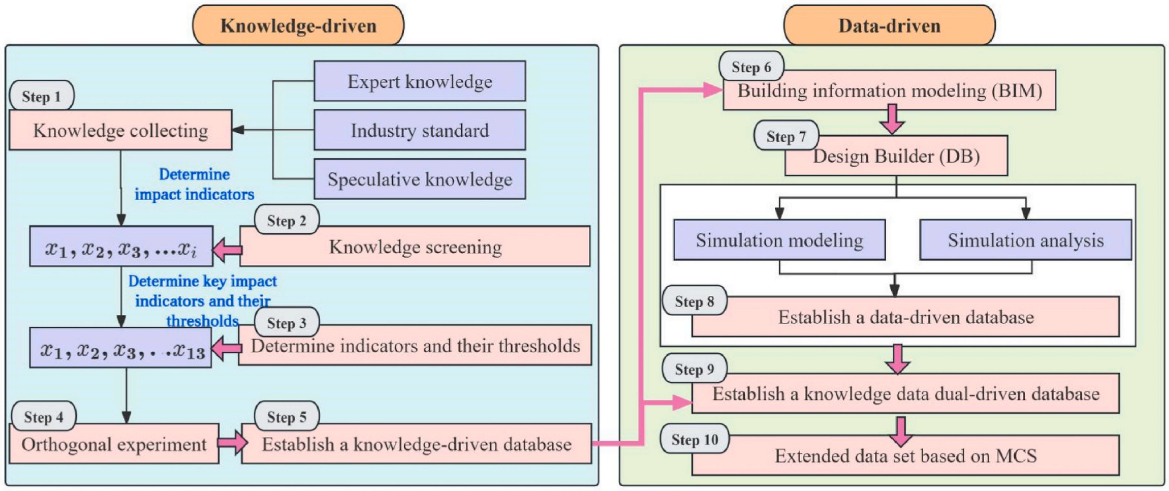


Fig. 2. A flowchart of the process of building a data set by the dual knowledge–data–driven model.

(MCS) is used to expand the dataset to increase the reliability. The CPO–NGBoost machine learning algorithm is employed to develop the model for predicting building performance, serving as the fitness function for parameter optimization. The MOGWO hybrid algorithm is used to obtain the Pareto frontier of multi-objective optimization. Ultimately, the ideal approach is utilized to determine the optimal configuration of the NZEB parameters.

3.1. Simulation calculation

3.1.1. Simulation parameter determination

We identify key indicators and threshold values through knowledge-driven methods, employ orthogonal experiments for systematic scheme design, and establish a knowledge-driven database to improve the efficiency and effectiveness of multi-indicator simulation analyses, thereby optimizing decision-making and research processes. Building on this foundation, the design schemes obtained from the knowledge-driven approach are incorporated into the BIM–DB for modeling and simulating public buildings across different climatic regions. The results generated from the simulation calculations are then synthesized to establish a sufficient sample data analysis repository for subsequent predictions and optimizations, providing a vital data foundation for this study. Fig. 2 presents a flowchart of the process of building a data set by the dual knowledge–data–driven model.

3.1.2. Orthogonal experiments

Orthogonal experimental design is a method of multi-factor and multilevel experimental design. This method can achieve the same results as one would achieve by conducting many comprehensive tests with the least number of tests possible; thus, the use of an orthogonal table design is an efficient, quick, and cost-effective method for performing multi-factor testing [55]. In this work, on the basis of the requirements of related specifications, Allpairs software [56] is used to obtain the orthogonal table to produce energy consumption datasets at various parameter levels for simulation.

3.2. CPO–NGBoost prediction model

3.2.1. Hyperparameter selection of CPO–NGBoost

In NGBoost, a reasonable selection of hyperparameters can reduce the error of the prediction results [24]. Therefore, before the NGBoost prediction model is established, it is necessary to select its hyperparameters effectively to improve the accuracy, robustness, and generalizability, among other performance parameters, of the model prediction process [25]. In this study, the hyperparameters that need to be adjusted are as follows: *n_estimators* is the number of trees, which is used to limit the tree algorithm from growing excessively deep; *learning_rate* controls shrinkage; and *colsample_bytree* refers to the column subsample ratio used during the construction of each tree that is used to regulate the complexity of the model.

3.2.2. Evaluation index

The prediction results are evaluated by the root mean square error (*RMSE*), mean absolute error (MAE), and goodness of fit (R^2) [57]. The smaller the model error is, the closer R^2 is to 1. The lower *RMSE* and *MAE* are, the better the performance of the model. R^2, *RMSE*, and *MAE* are calculated using Formulas (16), (17), and (18), respectively [58]:

$$R^2 = 1 - \frac{\sum_{i=1}^{n} \left(y^{obs} - y^{pred}\right)^2}{\sum_{i=1}^{n} \left(y^{obs} - \overline{y^{obs}}\right)^2} \tag{16}$$

$$RMSE = \sqrt{\frac{\sum_{i=1}^{n} \left(y^{obs} - y^{pred}\right)^2}{n}} \tag{17}$$

$$MAE = \frac{1}{n}\sum\nolimits_{i=1}^{n} \left|y^{obs} - y^{pred}\right| \tag{18}$$

where y^{obs} represents the corresponding data in the dataset and y^{pred} represents the results of the prediction model.

3.3. Multi-objective optimization

3.3.1. Selection of the objective function and the range of constraints

The predictive fitting relationship generated by the CPO–NGBoost algorithm serves as a fitness function in multi-objective optimization:

$$min\, f_m(x_1, x_2, \ldots, x_n) \tag{19}$$

where x_n refers to the decision parameter and f_m refers to the fitness function.

To avoid substantial differences between the real engineering parameters and the optimized values, it is important to set the range for the input parameters, as shown in Formula (20):

$$b_{il} \le x_i \le b_{iu} \tag{20}$$

where b_{il} is the lower bound, b_{iu} is the upper bound, and x_i is the input parameter.

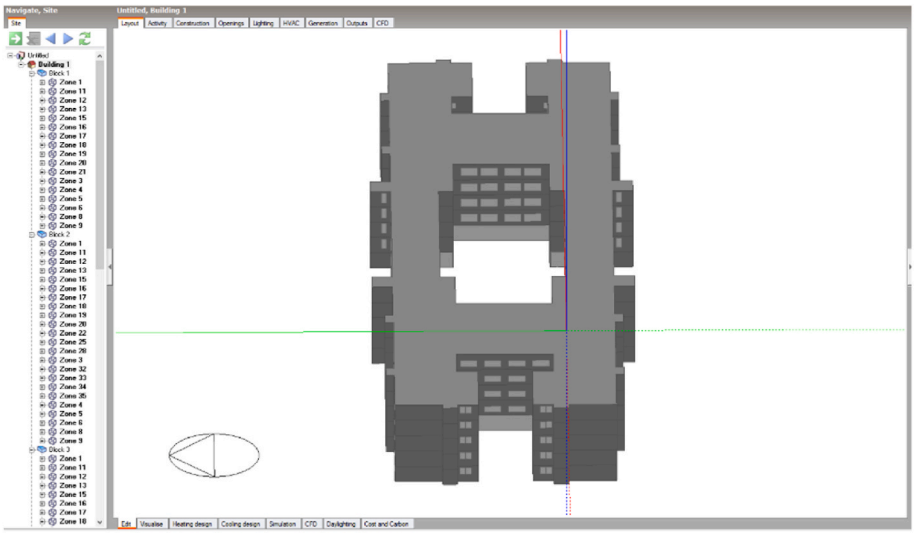


Fig. 3. Building performance simulation model of a five-story teaching building.

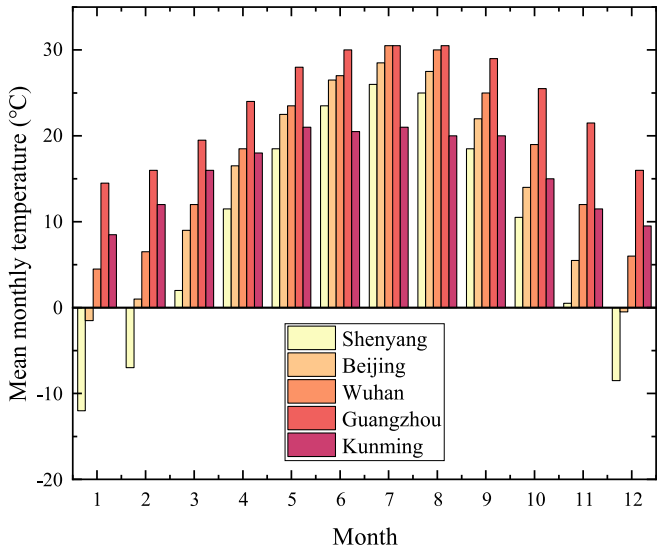


Fig. 4. Average daily temperatures in the five regions.

3.3.2. Ideal point method

The ideal point approach treats every option in multi-attribute decision-making as a point in multidimensional space and calculates the Euclidean distance between each scheme's point and the ideal point. The nearer a point is to the ideal point, the more advantageous the option is deemed to be. By considering four example objectives, the formula for calculating distance is as follows [59]:

$$U_n = \left(\frac{x_{pareto} - x_{Epoint}}{x_{Epoint}}\right)^2 + \left(\frac{y_{pareto} - y_{Epoint}}{y_{Epoint}}\right)^2 + \left(\frac{z_{pareto} - z_{Epoint}}{z_{Epoint}}\right)^2 + \left(\frac{w_{pareto} - w_{Epoint}}{w_{Epoint}}\right)^2 \quad (21)$$

where $(x_{E_{point}}, y_{E_{point}}, z_{E_{point}})$ is the coordinate of the ideal point and $(x_{pareto}, y_{pareto}, z_{pareto})$ is the coordinate of the Pareto frontier. Consequently, the aforementioned method can be used to identify a pair of optimal solutions within the set of Pareto-optimal solutions.

4. Case study

4.1. Project introduction

A university teaching facility in Wuhan, China, comprises five stories and measures 177 m long, 64.6 m wide, and 24 m tall. The building has a plot ratio of 1.21 and covers an area of 40,318 square meters. The classrooms in the building are symmetrically distributed, totaling 112 classrooms. Electricity serves as the sole energy source for the entire building, while a centralized air conditioning system manages the indoor temperature. The model of the teaching building was established in Revit software. The architectural floor plan was drawn, and room parameters (such as room names and areas) were set. Materials for building components (such as walls and roofs) were specified, and the thermal performance parameters of these components (such as the thermal transmittance coefficient) were set. The complete Revit model was then imported into DB software in the gbXML format, and building simulation parameters (such as climatic conditions) were set to conduct the simulation. As shown in Fig. 3.

The representative cities of China's five climate zones are Shenyang, Beijing, Wuhan, Guangzhou, and Kunming. Following the knowledge-driven index screening, the relevant climate data are imported into DB software to assess the impact of architectural design factors on building performance across various climatic scenarios. The average monthly temperature data of the five regions in 2021 are shown in Fig. 4.

4.2. BIM–DB building simulation

4.2.1. Selection of building parameters through knowledge-driven approaches

The BIM model is an effective method for simulating building performance by combining the unique material characteristics and external environmental conditions of buildings. [60]. Consequently, the BIM-integrated tool Design Builder is chosen for simulating BEC, CE, PMV, and GI.

Table 2
Parameter constraints of NZEBs via knowledge-driven methods.

Building design parameters	Extremely cold region	Cold region	Cold-winter and hot-summer region	Warm-winter and hot-summer region	Mild region
BOR (°)	0–90	0–90	0–90	0–90	0–90
EWHTC(W/(m^2·K))	0.1–0.25	0.1–0.3	0.15–0.4	0.3–0.8	0.2–0.8
EWSAC	0.4–0.7	0.4–0.7	0.4–0.7	0.4–0.7	0.4–0.7
RHTC (W/(m^2·K))	0.1–0.2	0.1–0.3	0.15–0.35	0.3–0.6	0.2–0.6
RSAC	0.4–0.7	0.4–0.7	0.4–0.7	0.4–0.7	0.4–0.7
AT (ac/h)	≤0.6	≤0.6	≤1	≤1	≤1
WWR	≤0.45	≤0.5	≤0.45	≤0.4	≤0.5
OWHTC (W/(m^2·K))	≤1.2	≤1.5	≤2.2	≤2.8	≤2.2
OWSAC	≤0.45	≤0.4	≤0.3	≤0.25	≤0.3
OWT	0.4–0.8	0.4–0.8	0.4–0.8	0.4–0.8	0.4–0.8
PD (person/m^2)	0.2–0.8	0.2–0.8	0.2–0.8	0.2–0.8	0.2–0.8
ACCT (°C)	20–26	20–26	20–26	20–26	20–26
ACHT (°C)	20–26	20–26	20–26	20–26	20–26

Before the energy consumption simulation, it is necessary to consider various factors that affect the analysis results, and basic parameters such as those influencing the building performance should be selected based on the knowledge about the actual items in use and their related specifications. For NZEBs, the architectural envelope parameters and the indoor environment both have considerable influence. Raji's research indicated that the energy usage of a building is largely determined by its envelope design elements [61]. Chen reported that enhancing air conditioning systems represents one of the most effective approaches to implementing NZEBs [62]. Mousavi's research revealed that proper air tightness design is an effective passive technique for improving NZEB performance [63]. Therefore, architectural envelope parameters and the indoor environment are selected as decision parameters. The selected architectural envelope parameters include the RSAC, RHTC, EWSAC, EWHTC, WWR, OWSAC, OWHTC, and OWT. The selected indoor environmental factors include AT, PD, ACHT, and ACCT. In addition, building orientation is an important factor affecting building performance. According to Al-Shargabi's research, the orientation of a building significantly influences its overall performance [64]. Therefore, BOR is selected as the basic parameter affecting building performance.

On the basis of the technical standards for NZEBs (GB/T 51350–2019), the general code for the utilization of renewable energy in building energy conservation (GB55015–2021), and other building envelope and energy system design requirements in China, the constraints of architectural design parameters in different climate zones are obtained, as shown in Table 2.

4.2.2. Results from orthogonal experiment and simulation

An orthogonal table with 13 factors and 7 levels is selected for each of the five regions by using Allpairs software, and BIM–DB is utilized to simulate 103 datasets. Table 3 shows the orthogonal experimental data in the mild region.

The results are calculated through DB simulation, and the BEC, CE, PMV, and GI levels are taken as the targets of the optimization step.

1. BEC

BEC includes the energy consumed by construction materials in the building process and the total energy utilized by electrical devices over a 50-year lifespan. For the convenience of calculation, it is assumed that BEC will remain unchanged during the 50 year service life. The method for determining BEC is presented as below:

$$BEC = BEC_m + BEC_y \times 50 \tag{22}$$

where BEC_m represents the energy consumption of construction materials in the building phase, while BEC_y represents the annual BEC during the operation period simulated by DB software.

2. CE

CE emissions from the manufacturing of building materials and cooling systems represent the largest share of overall carbon emissions [65,66]; however, when considering other greenhouse gas emissions, the CE simulation formula of this paper is as follows [67]:

$$CE = CE_{tee} + CE_{tce} + \theta \times BEC \tag{23}$$

where θ is the emission factor in the operation stage and CE_{tce} and CE_{tee} represent the carbon dioxide emissions and equivalent carbon dioxide emissions during the production phase, respectively.

3. PMV

The PMV is a holistic assessment index of human thermal comfort, formulated from the basic principles of human thermal balance and the levels of psychophysiological thermal sensation. The PMV index indicates the average voting index of seven levels of thermal sensation (+3∼-3), where a positive score signifies warmth, while a negative score denotes coolness. When the PMV approaches 0 more closely, the thermal environment becomes more comfortable.

4. GI

DesignBuilder employs the GI to evaluate visual discomfort at a reference point for the user. The GI acts as an index for predicting and evaluating visual discomfort caused by glare in interior working spaces, and its definition is as follows [23]:

Table 3
Orthogonal experimental data in the mild region.

Design parameters	Value level						
BOR (°)	0	15	30	45	60	75	90
EWHTC (W/(m^2·K))	0.2	0.3	0.4	0.5	0.6	0.7	0.8
EWSAC	0.4	0.45	0.5	0.55	0.6	0.65	0.7
RHTC (W/(m^2·K))	0.21	0.275	0.34	0.405	0.47	0.535	0.6
RSAC	0.4	0.45	0.5	0.55	0.6	0.65	0.7
AT (ac/h)	0.1	0.25	0.4	0.55	0.7	0.85	1
WWR	0.14	0.2	0.26	0.32	0.38	0.44	0.5
OWHTC (W/(m^2·K))	1	1.2	1.4	1.6	1.8	2	2.2
OWSAC	0.15	0.175	0.2	0.225	0.25	0.275	0.3
OWT	0.43	0.5	0.57	0.64	0.71	0.78	0.85
PD (person/m^2)	0.2	0.3	0.4	0.5	0.6	0.7	0.8
ACCT (°C)	20	21	22	23	24	25	26
ACHT (°C)	20	21	22	23	24	25	26

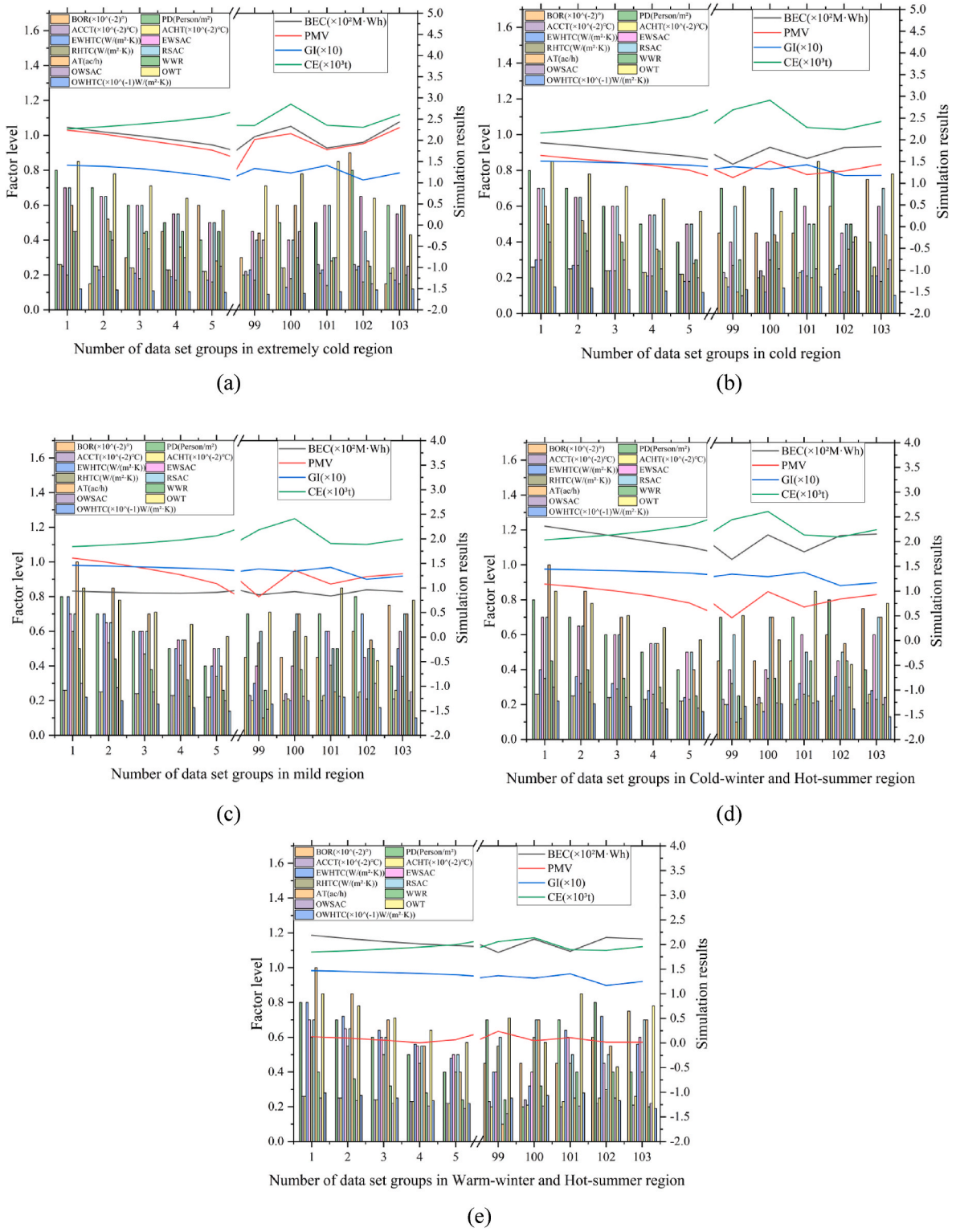


Fig. 5. Simulation results: (a) extremely cold region, (b) cold region, (c) cold-winter and hot-summer region, (d) warm-winter and hot-summer region, and (e) mild region.

$$GI = 10\,log_{10}\left(0.478\sum_{i=1}^{n}\frac{L_{si}^{1.6}\Omega_{si}^{1.8}}{L_b + 0.07\Omega_{si}L_{win}P_i^{1.6}}\right) \tag{24}$$

where L_b represents the background, L_{win} represents the window in cd/m^2, L_{si} denotes the luminance of the glare source, n represents the number of glare sources, Ω_{si} denotes the solid angle, and P_i denotes Guth's position index.

6. Building performance simulation results

The BEC, CE, PMV, and GI are simulated on the basis of the above 103 sets of data. Fig. 5(a–e) show 103 groups of simulation results in five regions of China. Architectural design parameters and the indoor

Table 4
Hyperparameter information of the NGBoost model.

Hyperparameters	Range
colsample_bytree	[0.4, 0.8]
learning_rate	[log(0.005), log(0.04)]
n_estimators	[80, 200]
minibatch_frac	[0.4, 0.8]

environment are very important to the sustainability and resource consumption of buildings. Under the same climate conditions, varying parameter settings have a significant impact on the simulation outcomes of building performance. For example, the maximum and minimum values of the BEC simulation results in the extremely cold region are 1543267.77 kWh and 25884.21 kWh, respectively, with a difference of 957616.44 kWh. Under different climate conditions, different architectural parameter designs have great impacts on the CE. The difference between the minimum and maximum CE simulation values is 1163.09 t. These findings highlight the critical importance of optimizing the architectural design parameters and of indoor environments of NZEBs.

4.3. CPO–NGBoost prediction

4.3.1. Dataset augmentation via Monte Carlo methods

In practice, the collection of large amounts of high-quality data is a difficult and tedious task. Owing to the limited amount of simulation data from real projects, Monte Carlo simulation (MCS) is adopted to simulate the random uncertainties in finite simulation data to accurately estimate the reliability of the simulation results of building performance [68]. Without considering the correlations between variables, the architectural design parameters and building performance follow a uniform distribution. The original 103 sets of simulation results and the MCS method are used to expand the data to 1010 sets.

4.3.2. Hyperparameter selection of the CPO–NGBoost algorithm

Prior to the development of the machine learning prediction model, it is very important to optimize and select the hyperparameters of the NGBoost algorithm to enhance the model's prediction performance. The hyperparameters and their definitions are provided in Table 4. The CPO hyperparameter search process takes the warm winter and hot summer region as an example, with BEC, CE, PMV, and GI as prediction targets,

Table 5
Hyperparameters of different prediction targets in the warm-winter and hot-summer region.

Hyperparameters	Prediction target			
	BEC	CE	PMV	GI
colsample_bytree	0.64	0.78	0.62	0.69
learning_rate	0.035	0.019	0.0049	0.024
n_estimators	140	100	180	140
minibatch_frac	0.61	0.58	0.77	0.62

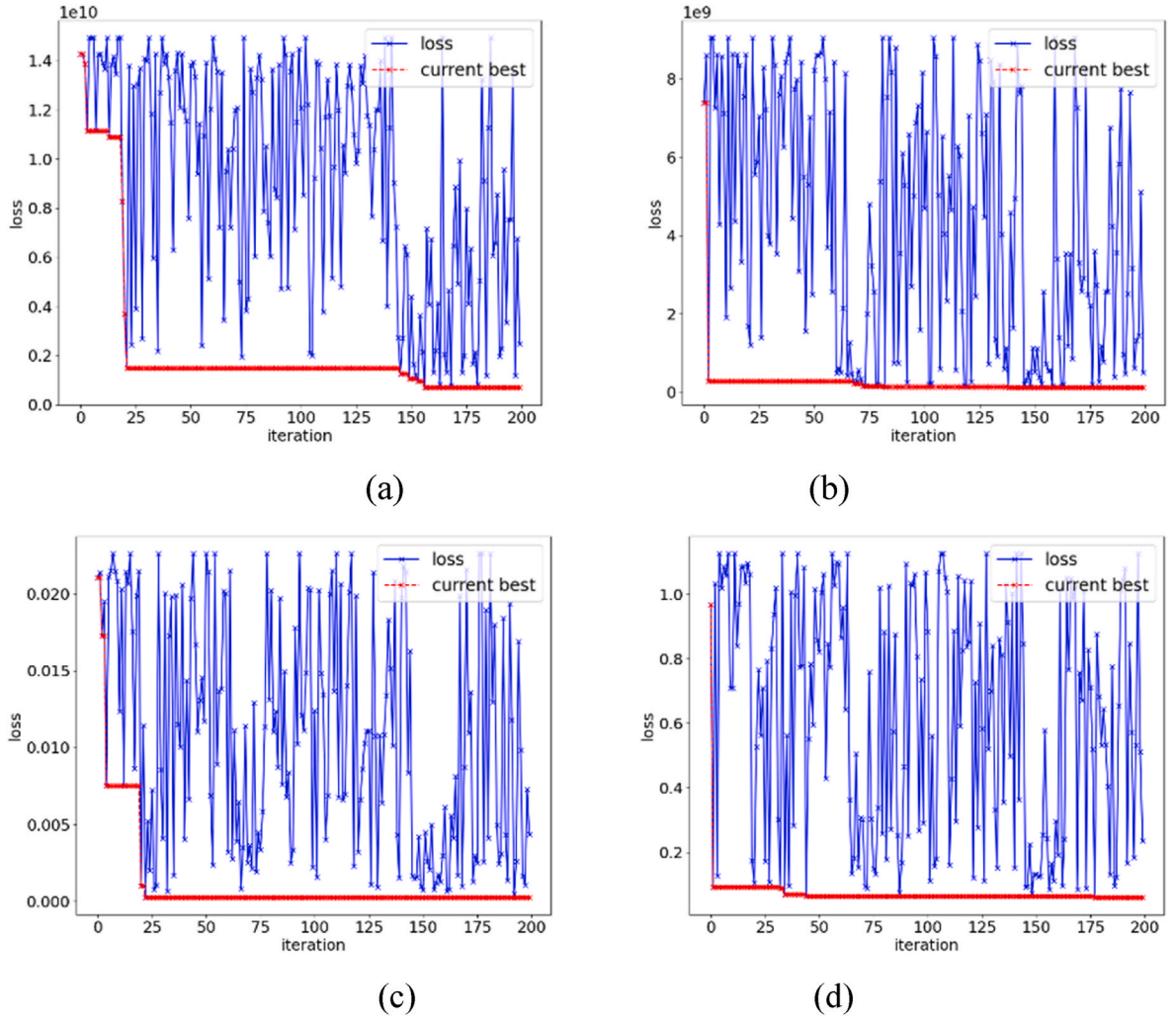


Fig. 6. Hyperparameter search procedure in the warm-winter and hot-summer region: (a) BEC, (b) CE, (c) PMV, and (d) GI.

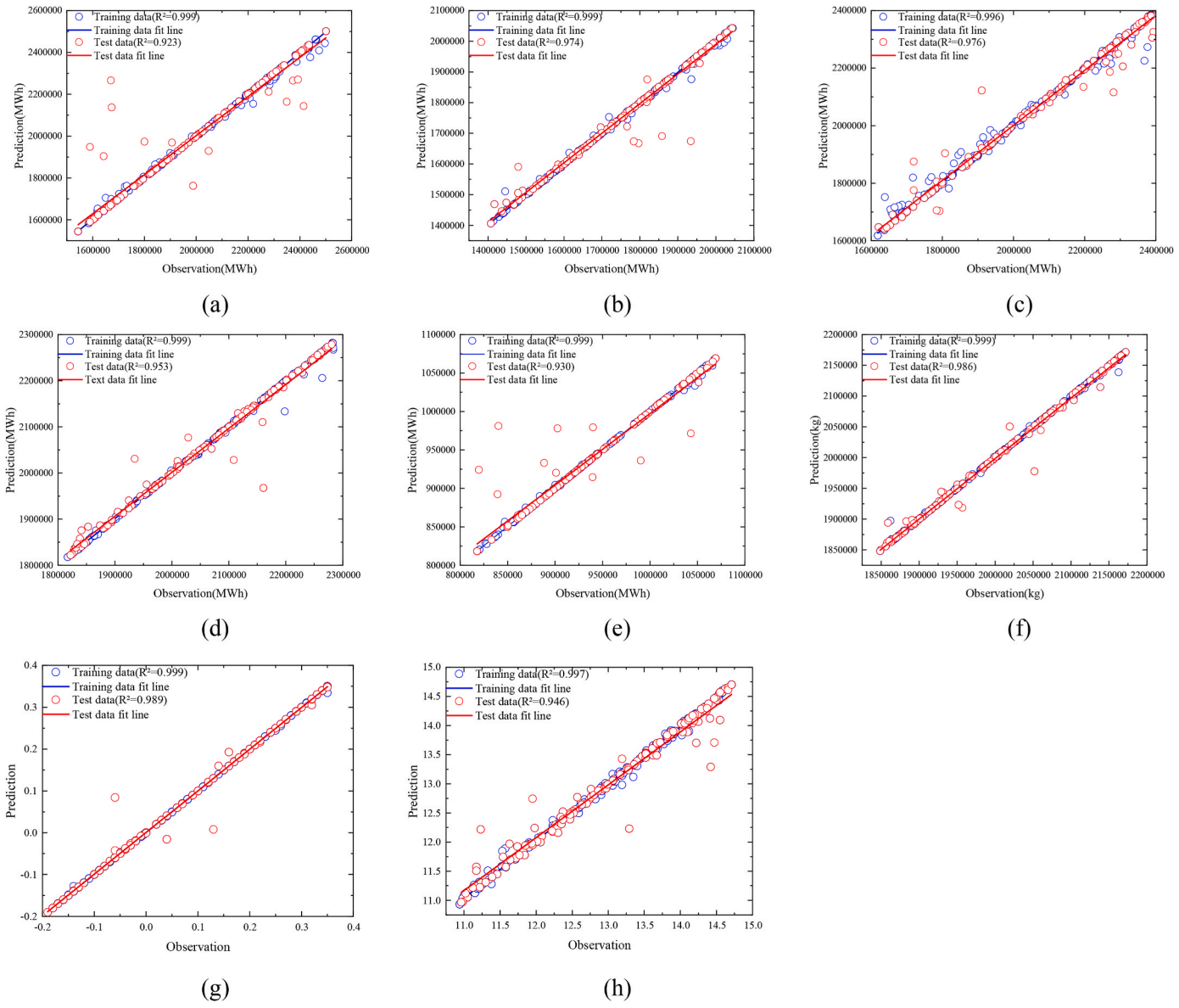


Fig. 7. Prediction renderings for the five regions: (a) BEC in the extremely cold region, (b) BEC in the cold region, (c) BEC in the cold-winter and hot-summer region, (d) BEC in the warm-winter and hot-summer region, (e) BEC in the warm region,(f) CE in the warm-winter and hot-summer region, (g) PMV in the warm-winter and hot-summer region, (h) GI in the warm-winter and hot-summer region.

Table 6
Constraint ranges of the parameters.

Design parameters	Constraint condition				
	Extremely cold region	Cold region	Cold-winter and hot-summer region	Warm-winter and hot-summer region	Mild region
BOR (°)	$0 \le x_1 \le 90$	$0 \le x_1 \le 90$	$0 \le x_1 \le 90$	$0 \le x_1 \le 90$	$0 \le x_1 \le 90$
EWHTC (W/(m^2·K))	$0.1 \le x_2 \le 0.25$	$0.1 \le x_2 \le 0.3$	$0.15 \le x_2 \le 0.4$	$0.3 \le x_2 \le 0.8$	$0.2 \le x_2 \le 0.8$
EWSAC	$0.4 \le x_3 \le 0.7$	$0.4 \le x_3 \le 0.7$	$0.4 \le x_3 \le 0.7$	$0.4 \le x_3 \le 0.7$	$0.4 \le x_3 \le 0.7$
RHTC (W/(m^2·K))	$0.1 \le x_4 \le 0.2$	$0.1 \le x_4 \le 0.3$	$0.15 \le x_4 \le 0.35$	$0.3 \le x_4 \le 0.6$	$0.2 \le x_4 \le 0.6$
RSAC	$0.4 \le x_5 \le 0.7$	$0.4 \le x_5 \le 0.7$	$0.4 \le x_5 \le 0.7$	$0.4 \le x_5 \le 0.7$	$0.4 \le x_5 \le 0.7$
AT (ac/h)	$0 \le x_6 \le 0.6$	$0 \le x_6 \le 0.6$	$0 \le x_6 \le 1.0$	$0 \le x_6 \le 1.0$	$0 \le x_6 \le 1.0$
WWR	$0 \le x_7 \le 0.45$	$0 \le x_7 \le 0.5$	$0 \le x_7 \le 0.45$	$0 \le x_7 \le 0.4$	$0 \le x_7 \le 0.5$
OWHTC (W/(m^2·K))	$0 \le x_8 \le 1.2$	$0 \le x_8 \le 1.5$	$0 \le x_8 \le 2.2$	$0 \le x_8 \le 2.8$	$0 \le x_8 \le 2.2$
OWSAC	$0 \le x_9 \le 0.45$	$0 \le x_9 \le 0.4$	$0 \le x_9 \le 0.3$	$0 \le x_9 \le 0.25$	$0 \le x_9 \le 0.3$
OWT	$0.4 \le x_{10} \le 0.8$	$0.4 \le x_{10} \le 0.8$	$0.4 \le x_{10} \le 0.8$	$0.4 \le x_{10} \le 0.8$	$0.4 \le x_{10} \le 0.8$
PD (person/m^2)	$0.2 \le x_{11} \le 0.8$	$0.2 \le x_{11} \le 0.8$	$0.2 \le x_{11} \le 0.8$	$0.2 \le x_{11} \le 0.8$	$0.2 \le x_{11} \le 0.8$
ACCT (°C)	$20 \le x_{12} \le 26$	$20 \le x_{12} \le 26$	$20 \le x_{12} \le 26$	$20 \le x_{12} \le 26$	$20 \le x_{12} \le 26$
ACHT (°C)	$20 \le x_{12} \le 26$	$20 \le x_{12} \le 26$	$20 \le x_{12} \le 26$	$20 \le x_{12} \le 26$	$20 \le x_{12} \le 26$

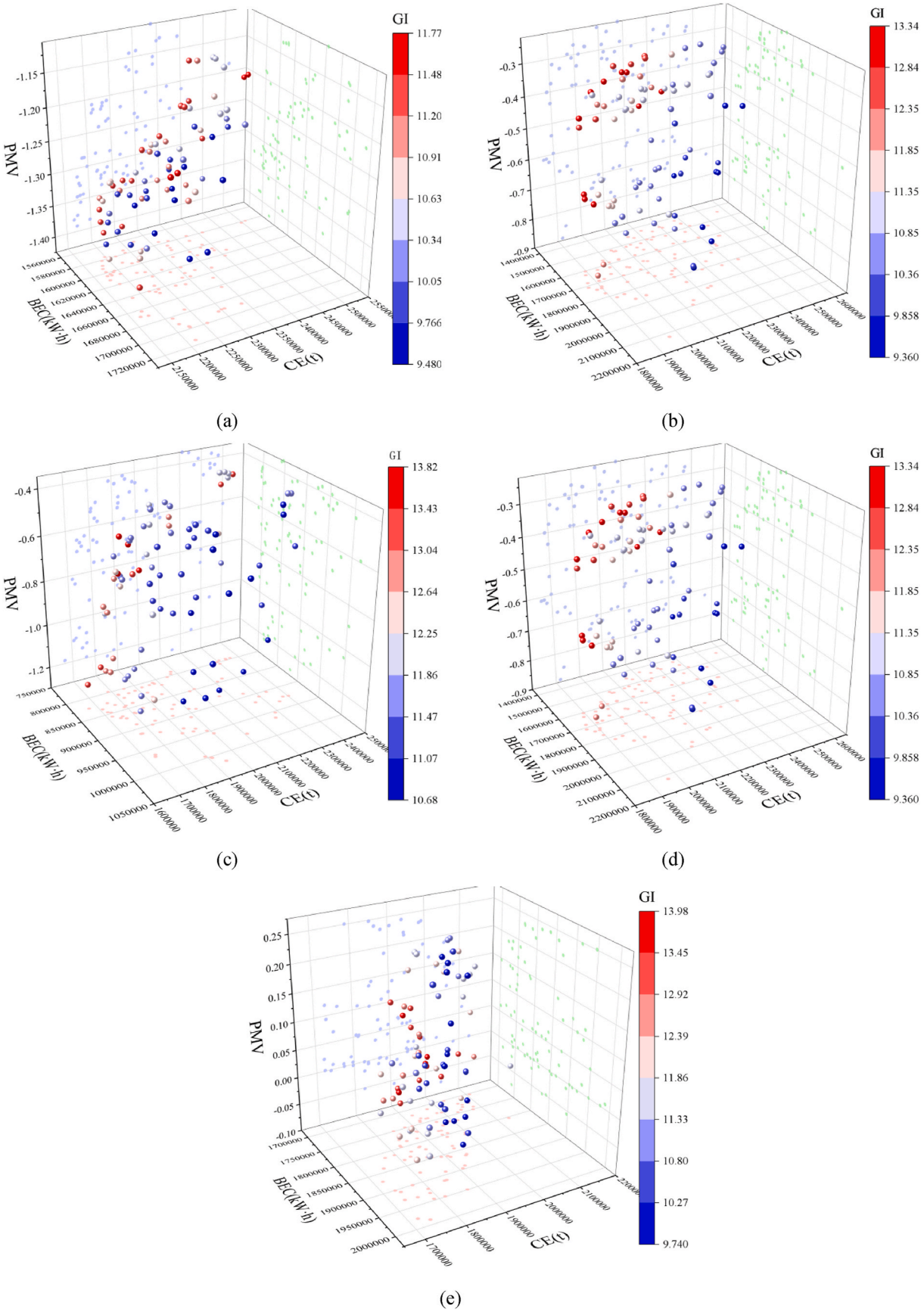


Fig. 8. Optimal Pareto front: (a) extremely cold region, (b) cold region, (c) cold-winter and hot-summer region, (d) warm-winter and hot-summer region, and (e) mild region.

Table 7
Comparison between IS and OS.

Building design parameters	Extremely cold region		Cold region		Cold-winter and hot-summer region		Warm-winter and hot-summer region		Warm region	
	IS	OS	IS	OS	IS	OS	IS	OS	IS	OS
BOR	0	75.56	0	0.43	0	6.32	0	9.20	0	11.15
EWHTC (W/(m^2·K))	0.65	0.23	0.65	0.27	0.65	0.39	0.65	0.80	0.65	0.79
EWSAC	0.55	0.54	0.55	0.55	0.55	0.67	0.55	0.43	0.55	0.65
RHTC (W/(m^2·K))	0.4	0.16	0.4	0.27	0.4	0.25	0.4	0.35	0.4	0.21
RSAC	0.35	0.54	0.35	0.42	0.35	0.60	0.35	0.56	0.35	0.65
AT (ac/h)	1	0.12	1	0.12	1	0.10	1	0.44	1	0.26
WWR	0.4	0.44	0.4	0.21	0.4	0.15	0.4	0.23	0.4	0.30
OWHTC (W/(m^2·K))	3	1.04	3	1.45	3	1.56	3	2.31	3	1.56
OWSAC	2	0.19	2	0.31	2	0.23	2	0.17	2	0.19
OWT	0.8	0.49	0.8	0.61	0.8	0.82	0.8	0.76	0.8	0.64
PD (person/m^2)	0.2	0.74	0.2	0.75	0.2	0.80	0.2	0.60	0.2	0.79
ACCT (°C)	24	21.24	24	22.23	24	22.20	24	23.09	24	26.00
ACHT (°C)	20	25.69	20	23.23	20	21.36	20	21.00	20	24.63
BEC (× 10^2 MWh)	24.35	15.93	20.44	14.41	23.64	16.03	22.83	19.01	10.02	8.25
Optimization rate	34.60 %		29.52 %		32.20 %		16.74 %		17.70 %	
CE (× 10^{2}t)	44.99	21.79	42.19	21.28	37.03	18.97	32.70	17.73	27.28	16.67
Optimization rate	51.61 %		49.57 %		48.77 %		45.77 %		38.91 %	
PMV	2.30	1.31	1.97	0.71	1.11	0.44	0.12	0.07	1.37	0.79
Optimization rate	42.90 %		63.91 %		60.69 %		43.80 %		42.63 %	
GI	12.33	10.94	13.12	10.99	12.80	10.82	13.16	11.87	13.52	11.05
Optimization rate	11.27 %		16.24 %		15.45 %		9.77 %		18.26 %	

Table 8
Recommended ranges of the NZEB design parameters.

No.	Parameters	Recommended value range				
		Extremely cold region	Cold region	Cold-winter and hot-summer region	Warm-winter and hot-summer region	Warm region
1	BOR (°)	60–80	0–75	0–60	10–90	10–90
2	EWHTC (W/(m^2·K))	0.21–0.25	0.15–0.30	0.16–0.40	0.36–0.80	0.20–0.80
3	EWSAC	0.40–0.69	0.51–0.58	0.40–0.69	0.43–0.62	0.44–0.65
4	RHTC (W/(m^2·K))	0.14–0.16	0.25–0.30	0.22–0.26	0.30–0.40	0.20–0.49
5	RSAC	0.40–0.58	0.41–0.64	0.40–0.63	0.44–0.68	0.42–0.70
6	AT (ac/h)	0.10–0.12	0.12–0.51	0.10–0.25	0.26–0.61	0.10–0.58
7	WWR	0.15–0.45	0.20–0.49	0.15–0.41	0.18–0.28	0.20–0.42
8	OWHTC (W/(m^2·K))	1.03–1.77	1.02–1.47	1.43–2.17	1.93–2.80	1.00–2.00
9	OWSAC	0.19–0.39	0.26–0.37	0.12–0.29	0.17–0.21	0.17–0.24
10	OWT	0.44–0.75	0.56–0.70	0.44–0.84	0.71–0.79	0.43–0.78
11	PD (person/m^2)	0.65–0.75	0.44–0.80	0.38–0.80	0.30–0.63	0.72–0.8
12	ACCT (°C)	20–24	22–24	21–25	21–25	20–26
13	ACHT (°C)	22–26	21–26	20–26	20–24	20–25

and 200 iterations, as shown in Fig. 6. The optimization results of the four superparametric targets (*colsample_bytree*, *learning_rate*, *n_estimators,* and *minibatch_frac*) selected in the NGboost model are obtained through superparametric optimization. Table 5 illustrates the specific results.

4.3.3. CPO–NGBoost prediction results

Following the process of training and fitting the dataset, the CPO-NGBoost model yields the building performance prediction. A visualization of the prediction results is shown in Fig. 7(a–h). The prediction outcomes indicate that the CPO–NGBoost model performs more effectively across various climatic conditions. In the context of this research, the ratio of the prediction set to the test set is 8:2. In the test set, the R^2 values for the CPO–NGBoost model exceed 0.9 across all predicted targets, which indicates that the predicted results have a good fit and that the actual building performance data have little error with respect to the predicted values. The actual building performance data are derived from the results of the BIM-DB simulation calculations. The *RMSE* and *MAE* values of the CPO–NGBoost model are very close to 0 for all the prediction targets, indicating that the prediction results have high stability and strong robustness. Previous studies have demonstrated the superiority of the NGBoost model in forecasting. Gong employed four machine learning algorithms to forecast return temperatures in district heating systems, and the NGBoost model demonstrated superior performance compared to the others concerning all typical assessment metrics [69]. Tavares used the NGBoost model to predict the future performance of metabolic syndrome and demonstrated its good predictive performance [70].

The CPO–NGBoost algorithm can adapt to the analysis of building parameters under different climatic conditions and obtain better prediction results. Most earlier research has concentrated on building performance in a specific climatic condition. Chen et al. employed the LSSVM machine learning algorithm to develop a predictive model for energy consumption in buildings located in hot-summer and cold-winter regions [6]. Deng simulated and analyzed the energy consumption of residential buildings in cold areas [71]. In this work, five climate zones in China are selected as test points for simulation and data analysis, verifying the comprehensiveness and extensiveness of the method. In addition, the good prediction results show that the CPO–NGBoost algorithm can effectively manage nonlinear data and complex models with high accuracy and stability. This finding proves that the CPO–NGBoost algorithm can obtain the correct fitness function for multiobjective optimization problems and improve the efficiency and accuracy of solving multiobjective optimization problems. Many studies have used ML predictive models to obtain fitness functions for multiobjective optimization problems [72].

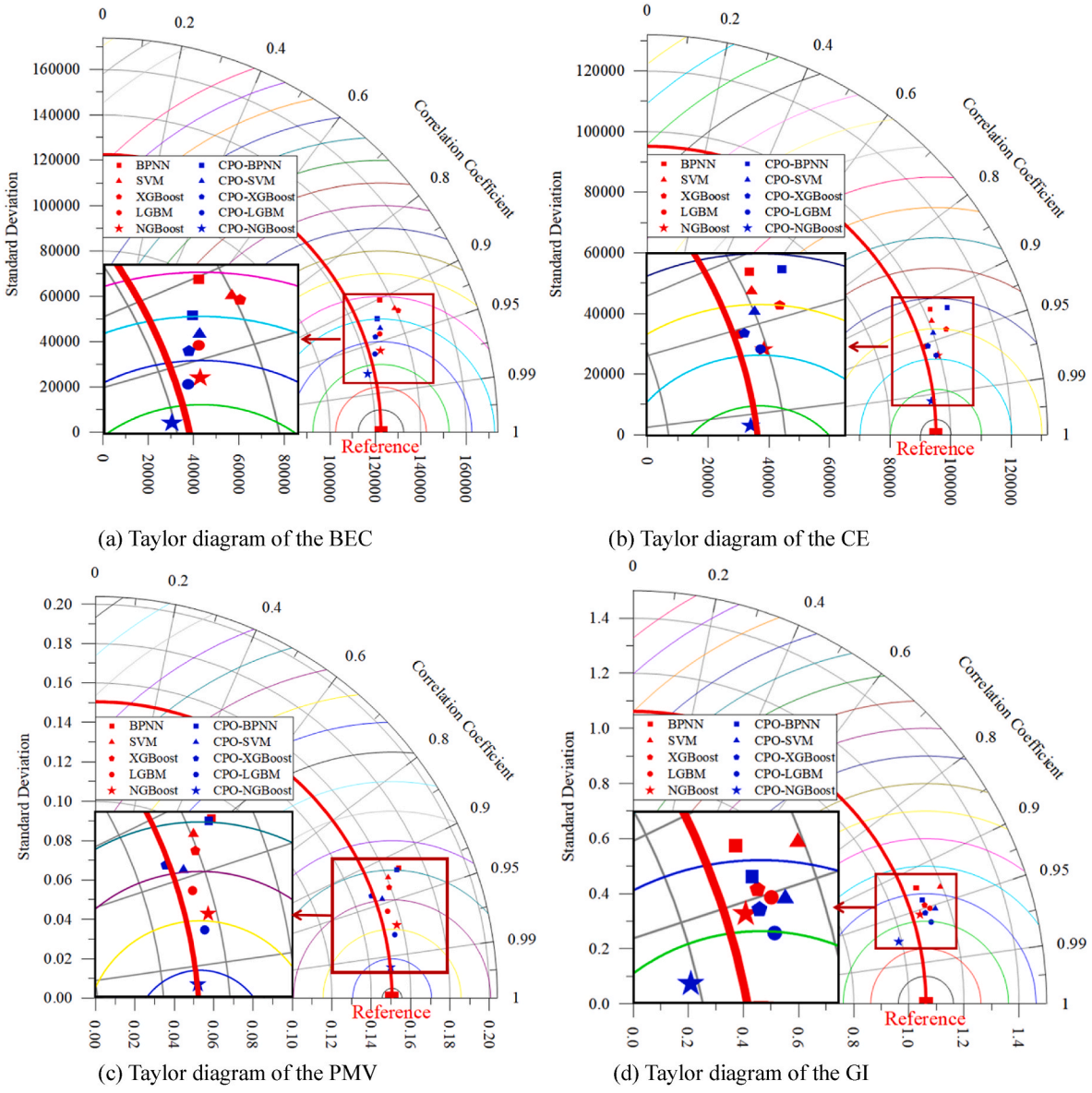


Fig. 9. Taylor diagrams of ten model predictions.

4.4. CPO–NGBoost–MOGWO optimization

4.4.1. Acquisition of the objective functions

As indicated in Section 3, the objective functions are acquired on the basis of the prediction model, as shown in Formula (25):

$$\begin{cases} f_1 = min\, f_{BEC}(\text{NGBoost } regression(x_1, x_2, x_3, x_4, x_5, x_6, x_7, x_8, x_9, x_{10}, x_{11}, x_{12}, x_{13})) \\ f_2 = min\, f_{CE}(\text{NGBoost } regression(x_1, x_2, x_3, x_4, x_5, x_6, x_7, x_8, x_9, x_{10}, x_{11}, x_{12}, x_{13})) \\ f_3 = min\, f_{PMV}|\text{NGBoost } regression(x_1, x_2, x_3, x_4, x_5, x_6, x_7, x_8, x_9, x_{10}, x_{11}, x_{12}, x_{13})| \\ f_4 = min\, f_{GI}(\text{NGBoost } regression(x_1, x_2, x_3, x_4, x_5, x_6, x_7, x_8, x_9, x_{10}, x_{11}, x_{12}, x_{13})) \end{cases} \tag{25}$$

4.4.2. Constraint condition determination

The value ranges of influencing factors in different climate zones are determined according to Table 3, as shown in Table 6.

4.4.3. Optimization results

CPO–NGBoost–MOGWO is used to optimize the values of BEC, CE, PMV, and GI for NZEBs across five distinct climatic zones. In the MOGWO code, the maximum evolutionary algebra is set to 100, the population is configured to consist of 100 individuals, and the target count for the GA is established at 4, and the population type of the CPO–NGBoost–MOGWO model is set to a bidirectional quantity. The MOGWO optimization algorithm is employed to obtain 84 optimal

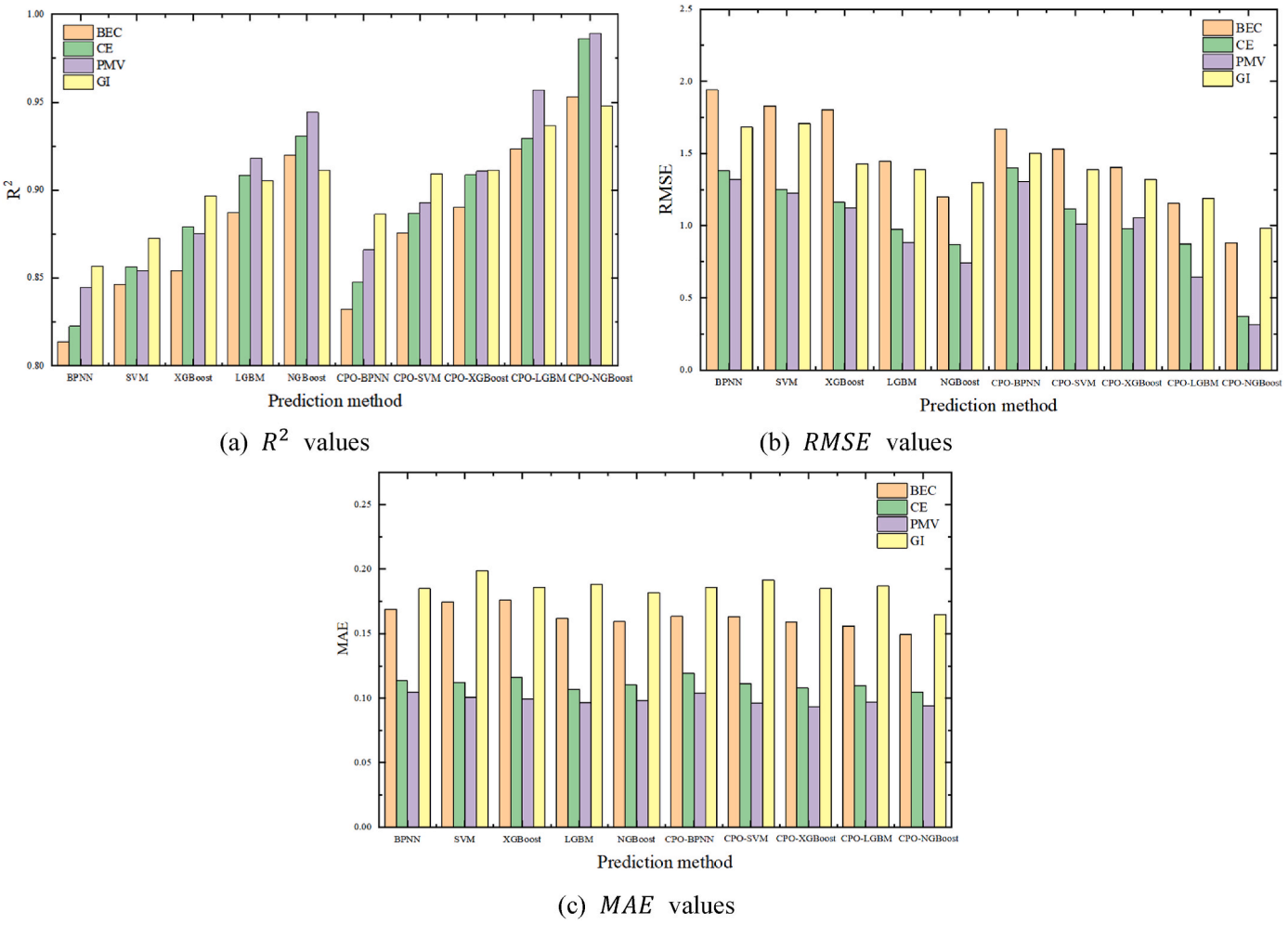


Fig. 10. Comparison of different models.

solutions in each climate region, forming Pareto boundaries that effectively balance multiple objectives or conditions. The distribution ranges of the Pareto frontier data can be seen in Fig. 8(a–e).

The minimum values of BEC, CE, PMV, and GI are chosen from the Pareto solution set to establish ideal points, and the distance from each point in this solution set to the ideal points is calculated using Formula (15). Ultimately, the scheme corresponding to the shortest distance is deemed the optimal one. The difference between the optimization scheme (OS) and the initial scheme (IS) is illustrated in Table 7. The results demonstrate that the intelligent algorithm and ideal point method employed in the optimization scheme can effectively enhance the performance of multiple buildings across different climatic conditions. The BECs in different climate zones are reduced by 16.74 %–34.60 %, the CEs are reduced by 38.91 %–51.61 %, the PMVs are reduced by 42.63 %–63.91 %, and the GIs are reduced by 9.77 %–18.26 %. In different climate environments, the sustainable development evaluation target values of buildings have been improved to varying degrees. The BEC and CE are greatly reduced as the living comfort of buildings is improved. Therefore, the intelligent optimization algorithm presented in this study effectively addresses the multi-objective optimization challenges related to building performance and supports the advancement of NZEBs across various climatic regions.

In addition, the optimization effect for the BEC varies in different climates. As indicated by the optimization results of the BEC, the optimization effects of extremely cold regions, cold regions, and cold-winter and hot-summer regions are good, with the optimization rate exceeding 29 %. However, the optimization rates of hot-summer and warm-winter areas and mild areas are less than 20 %. Owing to the relatively warm climates in warm-winter and hot-summer regions and warm regions, the energy demand is difficult to reduce via passive measures [50]. In this type of area, the energy-saving renovation of buildings is an effective way to further reduce BEC. For example, larger photovoltaic systems can be installed to achieve a given renewable energy utilization rate to reduce BEC. Zhang's research revealed that rooftop photovoltaics are critical for achieving an energy transition and have great energy saving and emission reduction potential [73]. The CE and PMV optimizations are good, with the CE and PMV optimizations exceeding 35 % and 40 %, respectively, in all climate zones.

Through statistical analysis of the NZEB design design variables within the Pareto-optimal solution set and combined with the actual situations of NZEBs in different regions, the suggested values for the NZEB design parameters are derived, as illustrated in Table 8. According to the reasonable range of values for building design parameters, parameters in line with local climate conditions can be formulated in the design stage, thereby achieving energy savings and emission reductions while promoting the development of NZEBs.

To meet the requirements of the optimization scheme of envelope structure parameters in different climate areas, it is necessary to integrate knowledge-driven scientific, reasonable, and effective measures to improve the sustainable development performance of public buildings in different climate regions. For example, according to standards such as the standard for materials for use in the external thermal insulation system of molded polystyrene board in thin plastered exterior walls (GB/T29906), it can be concluded that graphite polystyrene boards and

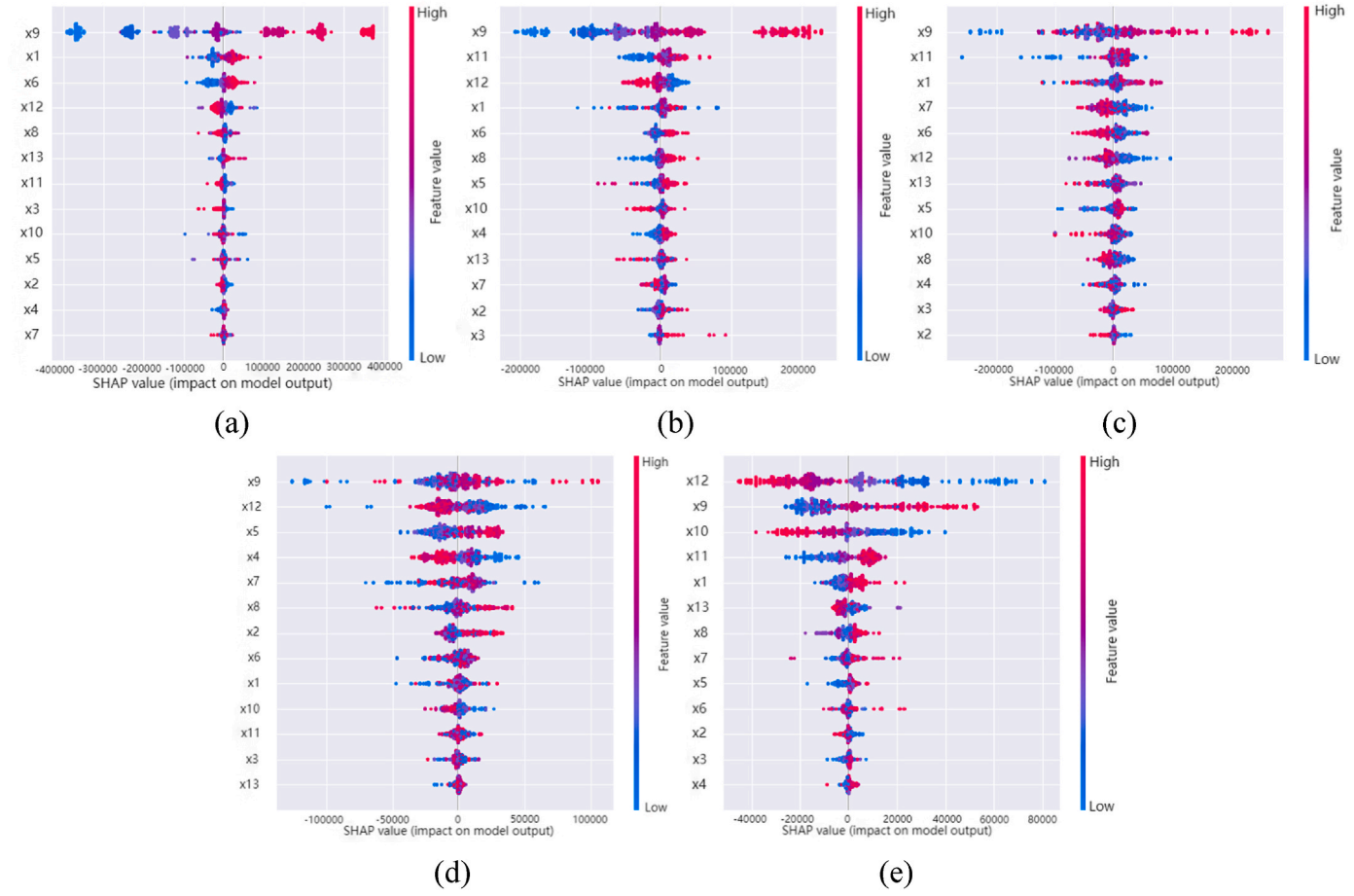


Fig. 11. Influences of building parameters on the BEC: (a) extremely cold region, (b) cold region, (c) cold-winter and hot-summer region, (d) warm-winter and hot-summer region and (e) mild region.

molded polystyrene boards are preferred as thermal insulation materials for exterior walls and roofs in extremely cold and cold regions, and aluminum alloy windows are selected for external windows in extremely cold regions. Moreover, double-layered insulating glass should be selected for external windows in cold zones. In regions with cold winters and hot summers, double-glazed hollow glass is recommended for external windows, and sunshade systems recommended.

5. Discussion

5.1. Comparison of the prediction results of different algorithms

In this work, BPNN, SVM and other common machine learning (ML) algorithms are selected to compare the prediction results with those of NGBoost. Moreover, to reflect the advantages of the CPO algorithm, the results of the ML algorithm with and without using CPO to optimize the hyperparameters are compared. By considering the area with hot summers and warm winters as an example, a Taylor diagram of the predicted results of each model is shown in Fig. 9.

The Taylor diagram visually shows the standard deviation, correlation coefficient, and root mean square error between the model's predicted values and the observed data, allowing for the evaluation and comparison of the performance of various prediction methods. By analyzing the position of the model on the Taylor chart illustrates metrics such as the distance from the origin to the model's prediction point, as well as the proportion of the model's predicted value standard deviation to that of the observed data. The distance from the observation data point to the model prediction point indicates the correlation coefficient between the model prediction value and the observation data. The Taylor diagram provides a quick visual representation of the model's accuracy, deviation, and correlation.

The diagram indicates that: the CPO–NGBoost algorithm is superior in target prediction in warm-winter and hot-summer regions. Among the four target forecasts, CPO–NGBoost has the best prediction effect and the highest prediction accuracy. In the test set results predicted by the CPO–NGBoost algorithm, R^2 ranges from 0.927 to 0.989, *RMSE* ranges from 0.314 to 1.495, and *MAE* ranges from 0.094 to 0.219, which shows that CPO–NGBoost has a good prediction effect. The good fitting effect indicates that CPO–NGBoost effectively addresses nonlinear data and complex models and has high accuracy and stability.

Furthermore, the prediction accuracy generally improves with the application of the CPO algorithm for hyperparameter optimization, highlighting the significance of tuning hyperparameters when utilizing the machine learning algorithm. The CPO approach is a precise and effective technique for hyperparameter optimization that can be applied in the tuning of the machine learning algorithm's hyperparameters [74]. The detailed prediction outcomes for the 10 models are presented in Fig. 10.

5.2. Explanatory analysis of influence index

The SHAP method is utilized to interpret and analyze the BEC prediction model and the PMV prediction based on CPO-NGBoost.

(1) The BEC prediction model

Fig. 11 highlights how building parameters affect Building Energy Consumption (BEC) predictions. In China, key factors like Building

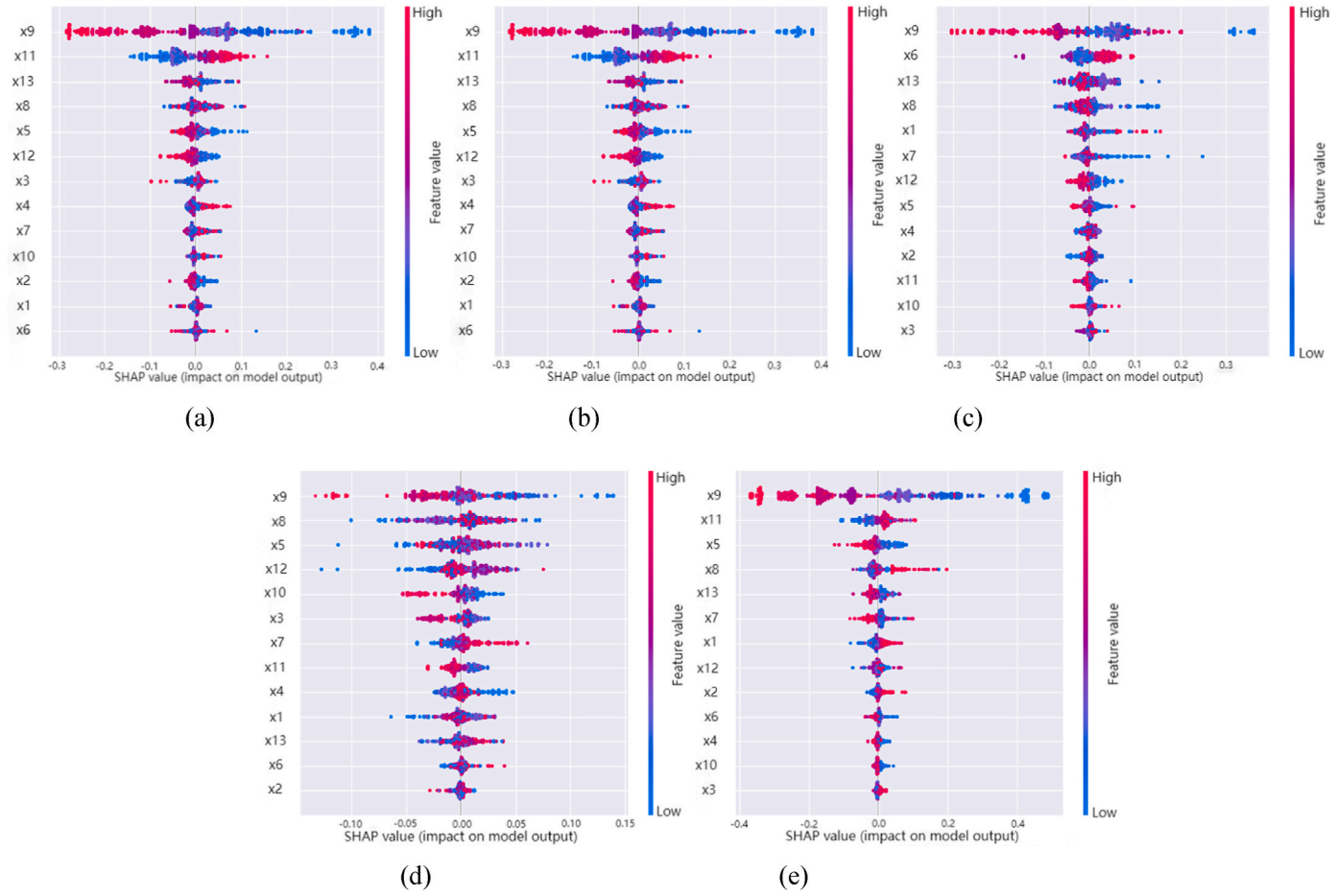


Fig. 12. Influences of building parameters on the PMV: (a) extremely cold region, (b) cold region, (c) cold-winter and hot-summer region, (d) warm-winter and hot-summer region, and (e) mild region.

Orientation (BOR), Overall Window-to-Wall Area Ratio (OWSAC), and Air Conditioning Set Temperature (ACCT) significantly influence BEC across different climates. Deng's research shows that adjusting BOR can reduce BEC in cold regions [71]. Kasalainen emphasizes the importance of window design for BEC [75], while Yan underscores the impact of ACCT on BEC [76]. Fig. 11 further clarifies these relationships: OWSAC positively correlates with BEC, meaning higher window area ratios increase energy use. Conversely, ACCT negatively correlates with BEC, so raising the set temperature lowers consumption. BOR's impact on BEC is complex and region-specific [77].

Building air tightness (AT) also plays a crucial role, especially in extremely cold, cold, and winter-cold summer-hot regions. Large indoor-outdoor temperature differences lead to heat loss and increased heating demand. Enhancing air tightness reduces this loss and is cost-effective. Teamah [78] and Tombarevic's [79] studies highlight its importance for BEC, particularly in cooler climates. In mild winters and hot summers or temperate zones, ACCT's impact on BEC is more pronounced due to high air conditioning use. Increasing ACCT can thus reduce BEC. Since Carbon Emissions (CE) calculation is related to BEC, optimizing parameters like BOR, OWSAC, ACCT, and AT can effectively lower BEC and achieve energy-saving goals [79].

(2) The PMV prediction model

Fig. 12 displays the effects of building parameters on the PMV prediction results. In different climate regions of China, the RSAC, OWHTC, and OWSAC have significant effects on the PMV prediction results. The RSAC and OWSAC exhibit a negative correlation with the PMV. The OWHTC and PMV in warm-winter and hot-summer regions and mild regions are positively correlated overall. In the other three areas, the PMV first increases with increasing OWHTC but decreases with increasing OWHTC once the PMV increases to a certain range. In summary, the thermal comfort of a building is strongly influenced by its envelope, and enhancing the envelope design presents considerable opportunities for lowering indoor energy consumption and increasing thermal comfort [80]. In addition, the contribution of the ACHT to PMV prediction is substantial in cold-winter and hot-summer regions, extremely cold regions, and cold regions but low in warm regions. In warm climates where the heating demand is low and the energy used for heating is significantly lower than that for cooling, the ACHT is less important than in other climates [81].

6. Conclusion

This paper presents an innovative multi-objective prediction and optimization method and analyzes its application in optimizing the performance of building sustainable development in different climate regions. BIM–DB software and the MCS method are combined to obtain the simulation parameters of building performance. The CPO–NGBoost machine learning algorithm is employed to develop the building performance prediction model and obtain the fitness function of the multi-objective optimization problem. Ultimately, the CPO–NGBoost–MOGWO hybrid algorithm, which employs a dual knowledge–data approach, is utilized. NZEB optimization schemes in different climate zones are obtained.

(1) Building performance data sets can be efficiently and accurately generated by establishing building models using BIM software and importing them into DB software in the gbXML format for simulation analysis, combined with the Monte Carlo Simulation (MCS) method. The integration of BIM-DB and MCS simplifies the data collection and processing procedures, reduces operational complexity, and effectively meets the demand for large amounts of data in building performance analysis. This method not only facilitates the acquisition of building performance data under different climatic conditions but also provides a solid theoretical and data support for subsequent research, demonstrating significant practical application value and broad applicability.

(2) A CPO-NGBoost performance prediction model has been established, in which CPO is used for multi-objective optimization of the NGBoost hyperparameters, addressing the issue of insufficient prediction accuracy in traditional models due to inaccurate hyperparameter selection. This approach not only enhances the model's adaptability and generalization when dealing with high-dimensional data but also achieves lightweight computation, reducing the time cost of analysis. The CPO-NGBoost model performs well in predicting building performance across different climatic regions, with the R^2 values in the training set ranging from 0.909 to 0.989, indicating strong prediction accuracy and generalization capability, making it suitable for building performance prediction in various scenarios.

(3) The MOGWO algorithm is utilized along with the findings from the CPO-NGBoost forecasting model to define the objective function and optimize the multi-objective challenge related to sustainable building performance, which includes building energy consumption (BEC), carbon emissions (CE), Predicted Mean Vote (PMV), and green index (GI). By simulating the hunting behavior of gray wolves and utilizing the group search mechanism, MOGWO effectively avoids local optima and quickly finds the global Pareto front solutions. Under different climatic conditions, this method shows high efficiency, stability, and diversity, significantly improving building performance. The optimization results indicate that BEC is reduced by 16.74 %–34.60 %, CE by 38.91 %–51.61 %, PMV by 42.63 %–63.91 %, and GI by 9.77 %–18.26 % across different climatic regions. Additionally, the SHAP method is used to conduct interpretability analysis of the prediction model, revealing the impact of key parameters on building performance in different climatic regions and providing specific guidance for practical applications.

The method proposed in this paper provides an effective solution for building sustainable development performance optimization in different climate regions by innovatively combining a variety of technologies and algorithms. By using the BIM-DB and MCS methods to efficiently obtain data, the CPO-NGBoost algorithm for accurate predictions, and the MOGWO hybrid algorithm to achieve multi-objective optimization, this method not only improves the performance of building sustainable development, but also indicates a new direction for future research. However, the method still has some limitations. For instance, while prioritizing the sustainability of buildings, it is equally important to evaluate ways to enhance their economic benefits. Although artificial intelligence provides convenience for people, it also generates carbon emissions [82]. Therefore, it is crucial to select more efficient models to reduce operational costs and environmental impact [83]. In addition, future research needs to take into account both the applicability and the economic viability of the methods [84]. Moreover, as this study only involves one type of building, future research should cover a wider range of building types to enhance the generalizability of the method.

CRediT authorship contribution statement

Hongyu Chen: Formal analysis, Writing – original draft, Data curation. **Geoffrey Qiping Shen:** Conceptualization, Project administration. **Xinyi Li:** Investigation, Methodology. **Wen Xu:** Software, Supervision. **Yang Liu:** Writing – review & editing, Funding acquisition, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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